

Yiyang Su

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Education

Michigan State University

PH.D. IN COMPUTER SCIENCE AND ENGINEERING

- Advised by Dr. Xiaoming Liu
- GPA: 3.6/4.0

East Lansing, MI

Aug. 2021 - present

University of Rochester

B.S. IN COMPUTER SCIENCE AND B.S. IN MATHEMATICS

- Highest Distinction in Computer Science, High Honors in Research in Computer Science
- Highest Distinction in Mathematics, Honors in Research in Mathematics
- GPA: 3.96/4.0; Magna Cum Laude; Dean's List for all eligible semesters

Rochester, NY

Aug. 2017 - May 2021

Honors & Awards

- | | | |
|------|---|---------------------------|
| 2021 | Engineering Distinguished Scholar , 1 year of support awarded to outstanding incoming PhD students | Michigan State University |
| 2021 | Phi Beta Kappa , elected member through the Iota of New York Chapter | University of Rochester |
| 2021 | Donald M. Barnard Prize , awarded to 10 junior/senior engineering students based on merit | University of Rochester |

Publications

- [1] **Yiyang Su**, Minchul Kim, Feng Liu, and Xiaoming Liu. "Improving Open-Set Biometrics with Deep Nearest Neighbors". *In submission*. 2025.
- [2] Feng Liu, Minchul Kim, Zhiyuan Ren, **Yiyang Su**, Zhangyang Wang, Humphrey Shi, Anil K. Jain, and Xiaoming Liu. "Uncovering the Disentanglement Capability in Diffusion Models for Human Image Synthesis and Editing". *In submission*. 2025.
- [3] **Yiyang Su**, Feng Liu, and Xiaoming Liu. "Hierarchical and Adaptive Mixture of Biometric Experts for Video-based Person ReID". *In submission*. 2025.
- [4] **Yiyang Su**, Minchul Kim, Feng Liu, and Xiaoming Liu. "3D Motion-Aware Gait Recognition via Auto-Regressive Human Body Modeling". *In submission*. 2025.
- [5] **Yiyang Su**, Minchul Kim, Feng Liu, Anil K. Jain, and Xiaoming Liu. "Open-Set Biometrics: Beyond Good Closed-Set Models". *Proceedings of the European Conference on Computer Vision (ECCV)*. 2024.
- [6] Minchul Kim, Feng Liu, **Yiyang Su**, Anil K. Jain, and Xiaoming Liu. "KeyPoint Relative Position Encoding for Face Recognition". *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024.
- [7] Feng Liu, Ryan Ashbaugh, Nicholas Chimitt, Najmul Hassan, Ali Hassani, Ajay Jaiswal, Minchul Kim, Zhiyuan Mao, Christopher Perry, Zhiyuan Ren, **Yiyang Su**, Pegah Varghaei, Kai Wang, Xingguang Zhang, Stanley Chan, Arun Ross, Humphrey Shi, Zhangyang Wang, Anil Jain, and Xiaoming Liu. "FarSight: A Physics-Driven Whole-Body Biometric System at Large Distance and Altitude". *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2024.
- [8] Zhiyuan Ren, **Yiyang Su**, and Xiaoming Liu. "ChatGPT-Powered Hierarchical Comparisons for Image Classification". *Advances in Neural Information Processing Systems (NeurIPS)*. 2023.
- [9] **Yiyang Su**, Ali Vosoughi, Shijian Deng, Yapeng Tian, and Chenliang Xu. "Separating Invisible Sounds toward Audio-Visual Scene-Aware Separation". *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*. 2023.

Professional Experience

Research Assistant

Michigan State University

COMPUTER VISION LAB

Aug. 2021 - present

- Developed a novel loss function aligned with the objectives of open-set biometrics to address the misalignment of existing loss functions. Applied this loss function to face/gait recognition and person reID and achieved an improvement of up to 2.9% on closed-set and 4.2% on open-set performance. This work has been accepted to ECCV 2024 [5].
- Proposed a novel, training-free scoring algorithm for biometric models by combining original model scores and K-Nearest Neighbors scores which assess the presence of each query in the gallery. Conducted a comprehensive analysis of the algorithm's behavior using synthetic datasets, supported by theoretical justifications of its performance. Experimental results demonstrated up to a 10% improvement in both verification and open-set search. Prepared a research paper on this work, which is currently under submission [1].
- Created a video-based person reID framework that independently models appearance, body shape, and pose features and adaptively integrate them using a hierarchical mixture of expert model based on the input. This method improved the accuracy by up to 11%. Prepared a submission [3].
- Introduced future 3D gait sequence inference as an auxiliary task to learn 3D-aware spatial and temporal gait representations. Validated the effectiveness of the auxiliary task on both indoor and outdoor gait datasets, achieving up to 3.4% performance improvement in recognition accuracy. Prepared a first-author submission on this work [4].
- Developed a novel score normalization technique to semantically align the mean of genuine scores and the mean of imposter scores across different models. This approach achieved a 19% total improvement across four key evaluation metrics.
- Collaborated on the development, debugging, and maintenance of a complex, end-to-end system integrating detection, restoration, and biometric modules for the 4-year IARPA-funded BRIAR project with \$12.5M in funding. Trained gait recognition models on the BRIAR dataset and successfully deployed them as containerized modules within the integrated system.

Research Intern

University of Rochester

DR. CHENLIANG XU'S GROUP

May 2020 - May, 2021

- Formulated the novel problem of separating invisible sounds from sound mixtures, which existing audio-visual sound separation methods cannot address. Designed a model that leverages video semantics to separate all sounds, with and without visible sources. Conducted extensive experiments to validate the model's effectiveness and summarized the results in a research paper. [9].

Workshop Leader

TRIPODS NSF REU/GRADSTEMFORALL 2020

University of Rochester

Aug. 2020

- Co-led Python workshops to introduce foundational programming concepts, with a focus on applying Python to visualize and probabilistically verify theorems in high-dimensional probability.
- Developed and demonstrated Python scripts to simplify complex mathematical concepts through hands-on examples to enhance participants' understanding of theoretical proofs.

Research Intern

UNDER THE SUPERVISION OF DRS. YAPENG TIAN AND CHENLIANG XU

University of Rochester

Dec 2017 - Feb 2018

- Constructed a dataset of 10-second videos of 10 musical instruments with about 60 videos for each instrument.
- Used a pre-trained inception model to extract feature representations from each video and trained a classifier using these extracted features.

Research Intern

RSI-TSINGHUA SUMMER PROGRAM

Tsinghua University

Jul. 2016 - Aug. 2016

- Developed a convolutional neural network-based system to detect distracted drivers by analyzing visual cues of inattention.

Teaching and Mentorship

Mentorship

COMPUTER VISION LAB

Michigan State University

Sep. 2022 - present

- Detection & Tracking (MSU UG): Hank Murdock (Fall 2022), Apar Mohabansi (Fall 2022), Manh Tran (Fall 2022), Calvin DeJong (Fall 2023), Duc Hung Luong (Fall 2023)
- 3D Scene Reconstruction, Human Parsing, Person reID, etc: ZiAng Gu (MSU UG, Fall 2022 - present)
- MSU HSHSP: Zoaib Sihorwala (Summer 2023)

Teaching Assistant

VARIOUS DEPARTMENTS

University of Rochester

Sep. 2018 - Dec. 2020

- Functions of Real Variable I (Honors) (MATH 265H, Fall 2020)
- Intermediate Microeconomics (ECON 207, Fall 2020)
- Walk in CS tutoring (Spring 2020)
- Design and Analysis of Efficient Algorithms (CSC 282, Fall 2019)
- Computer Organization (CSC 252, Spring 2019)
- Computation and Formal Systems (CSC 173, Fall 2018)

Services

Server Manager

COMPUTER VISION LAB

Michigan State University

Apr. 2022 - present

- Oversee the management of 86 GPUs across 13 servers and ensure optimal performance and resource allocation for computational tasks.
- Coordinate with IT and support staff to maintain, update, and troubleshoot servers, as well as to plan and execute the procurement of new hardware.

Reviewer

FOR VARIOUS CONFERENCES

May 2022 - present

- FG 2025, WACV 2025, IJCBLRR 2024, ECCV 2024, ECCV 2022

Skills

Programming

Proficiency in Python, Java, C/C++, $\text{\LaTeX}$

Development

Familiarity with PyTorch and TensorFlow; working knowledge of Linux and command line tools

Languages

Native Chinese (Mandarin/Cantonese), fluent English, and elementary Japanese & French